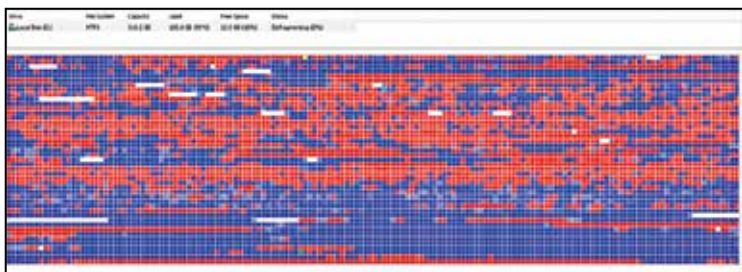


allows as many additional downloads as possible to happen on the extra drive without hindering the recovery process.

Right at the end of this chapter, we've given you a list of some free recovery software that we think should do the job for you. In the meanwhile, let's move on the next section of this chapter.

What is the efficiency of the recovery process?

Many times upon trying to recover some data, you may find that very few of the lost files will be 100 per cent recoverable. Why is that the case? Why are only parts of the data recoverable? Will it be usable? Well, the answers to these questions lie in a fact we've already mentioned previously. We spoke about how data is stored on your drive in clusters – but the thing is, these clusters aren't necessarily all stored next to each other and in one place. They're usually stored in different parts – this is called 'fragmentation'.



So many sectors so little time

Even the smallest of files could be heavily fragmented and spread over a hundred different parts of a hard drive.

If you've ever defragmented your disk, you'd already be familiar with the concept. In case you're not, Windows users could head over to Start > System and security > Administrative tools > Disk Defragmenter > Select drive > Defragment Disk to view its graphical representation.

So, when the computer sees a bit of space that could be overwritten, say, the space occupied by only 10 per cent of a particular file, with the remaining 90 per cent in parts elsewhere, it allows a new file to take up that space. This is what prevents recovery software from successfully retrieving some files in their entirety. A file's usability depends on the type of file it is and the parts of the file that are missing. However, recovering a file that's had its data overwritten on by a new file's data usually proves futile.